Huilin Song (宋惠霖)

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songhl.github.io

Education

Sun Yat-sen University, Guangdong, China 2022.9 – 2026.6

BSc in Information and Computing Science | GPA: 3.9/5.0 (89/100)

The Hong Kong Polytechnic University, Hong Kong, China 2026.9 – 2030.8 (expected)

PhD in Applied Mathematics | Research Focus: Financial Mathematics

English Proficiency

CET-6: 590/710 （Listening 205/248，Reading 215/248，Writing 170/214）
IELTS: 6.5 (Listening 6，Reading 7.5，Writing 6.5, Speaking 6.5)

Research Experience

Sun Yat-sen University(SYSU), China 2024.9 – 2024.12

Responsible for coding and main sections of paper writing

Brief introduction: This research focuses on improving the understanding capabilities of multimodal video-language large models (VLLMs) using adaptive sparse memory for long videos.

- Under the guidance of Jisheng Dang (PhD student at SYSU), we introduced a cross-attention－based Memory Bank module to enhance the model’s understanding of long-video contexts and improve visual–text alignment.
- Experimental results demonstrated significant improvements over the baseline in various video-question answering scenarios.
- The paper has been accepted by ICME 2025.

Sun Yat-sen University, China 2024.12 – 2025.3

Responsible for discussion and determination of core innovations, coding, and main sections of paper writing

Brief introduction: This research focuses on addressing single-frame bias in long videos and aims to reduce hallucinations in VLLMs for long video temporal understanding tasks while improving model performance and computational efficiency.

- Identified and validated the existence of single-frame static bias in long video understanding tasks by proposing the DTP module.
- Modified the MA-LLM Memory Bank module to enable the discrimination of static bias and dynamic selection based on it, and incorporated a CLIP-based contrastive learning module to enhance visual-text alignment.

- Experimental results demonstrated improvements over the baseline in various video question answering and video captioning tasks, with significantly reduced computation time.

Sun Yat-sen University, National University of Singapore (NUS) 2025.4 – 2025.8

Responsible for conception, algorithm design, coding, and manuscript writing

Brief introduction: In this work, we introduce MUPA, a collaborative multi-agent framework that seamlessly integrates video grounding, question answering, self-reflection, and evidence aggregation to enhance the trustworthiness and reliability of multimodal LLMs for Grounded VideoQA.

- Designed three complementary reasoning paths and a reflection agent (PoE + MoE fusion) that jointly boost grounding fidelity *without* sacrificing answer accuracy.
- Achieved new state-of-the-art on NExT-GQA and DeVE-QA. Remarkably, the 2B variant already surpasses all existing 7B baselines.
- Supervised by Junbin Xiao (NUS Postdoctoral Fellow) and Jisheng Dang (SYSU).
- The paper has been submitted to IJCV.

Publications and Manuscripts

** indicates equal contribution; † indicates corresponding author.*

- Bimei Wang*, **Huilin Song***, Jisheng Dang†, et al. "AS Memory: Adaptive Sparse Memory Meeting Video Language Models," accepted by *ICME 2025*.

Personal Skills

- **Solid Mathematical Foundation**
  Completed courses including Mathematical Analysis, Linear Algebra, and Optimization Theory, achieving scores above 90 in most courses.
- **Strong Computer Science Background**
  Proficient in C++, Python, and Linux; completed foundational courses such as Data Structures, Computer Networks and Parallel Algorithms. Experienced with mainstream deep learning frameworks.
- **Research Competence and Teamwork**
  Capable of independent literature review, experimental design, and fluent in reading and writing academic papers in English. Demonstrated strong teamwork and communication skills through collaborative research experiences, effectively cooperating with supervisors, PhD students, and peers.